# Conditional Probability

## ECON 97: Economic Toolkit

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Spring 2026

# iClicker course link

https://join.iclicker.com/RDNG

# Quiz: Sets

Draw three cards simultaneously from a standard deck of 52 playing cards. What is the probability of observing at least one king?

A. 0.1356

B. 0.1546

C. 0.2174

D. 0.3216

# Agenda

1. Recap of material
2. Conditional probability
3. Independence
4. Coding recap
5. Discrete R.V.

# Review of Concepts - Week 1

<u>**Important formulas**</u>:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Compliment: $P(A') = 1 - P(A)$
- Mutually exclusive: $P(A \cap B) = 0$
- Exhaustive: $P(A \cup B) = 1$
- $P(A \cup B)' = P(A' \cap B')$ and $P(A \cap B)' = P(A' \cup B')$
- $P(A \cap B') = P(A) - P(A \cap B)$
- Subset: if $A \subset B$ then $P(A \cap B) = P(A)$

**Important formulas**:

- Replacement, order matters: $n^k$
- No replacement
  - Order matters: ${}_nP_k = \frac{n!}{(n-k)!}$
  - Order doesn't matter: ${}_nC_k = \frac{n!}{(n-k)!k!}$
- ${}_nC_k = {}_nC_{(n-k)}$
- ${}_nC_n = 1$
- ${}_nC_1 = n$
- ${}_nC_0 = 1$

# Conditional Probability

We often care about multiple events at once.

Sometimes the probability of once event depends on whether another event has occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

E.g., probability that a patient has a disease given that a medical test is positive.

# Conditional Probability - Table

Suppose a certain medical test is used to diagnose a disease among 1000 patients, but the test is not 100% accurate.

| | Positive Test $(B_1)$ | Negative Test $(B_2)$ | Total |
|---|---|---|---|
| Has Disease $(A_1)$ | $45$ | $5$ | $50$ |
| No Disease $(A_2)$ | $90$ | $860$ | $950$ |
| Total | $135$ | $865$ | $1000$ |

# Table Probabilities

$$P(\text{Disease}) = \frac{50}{1000} = 0.05 \qquad P(\text{Positive}) = \frac{135}{1000} = 0.135$$

$$P(\text{Positive} \mid \text{Disease}) = \frac{45}{50} = 0.90$$

$$P(\text{Negative} \mid \text{No Disease}) = \frac{860}{950} \approx 0.905$$

$$P(\text{Disease} \mid \text{Positive}) = \frac{45}{135} = \frac{1}{3} \approx 0.333$$

$$P(\text{No Disease} \mid \text{Positive}) = \frac{90}{135} = \frac{2}{3}$$

$$P(\text{Disease} \mid \text{Negative}) = \frac{5}{865} \approx 0.0058$$

# Practice

An urn contains 6 colored balls: 2 orange and 4 blue. Two balls are selected at random without replacement. Suppose that you are told that at least one of them is blue. What is the probability that the other ball is also blue?

# Independence

If the probability of one event does not depend on the probability of another (and vice versa), then these events are **independent**.

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

$$P(A \cap B) = P(A)P(B)$$

If two events are independent, so are their complements.

Independence can extend to three or even more events (A, B, C, etc.)

# Practice

An urn contains 2 red balls and 4 white balls. Sample successively 5 times at random and with replacement so that the trials are independent. Compute the probability of each of the 2 sequences WWRWR and RWWWR.

# Practice

A power utility can supply electricity to a city from 3 different power plants. Suppose that each power plant fails with probability 0.2, independent of the others. Suppose that any one plant can provide enough electricity to supply the entire city. What is the probability that the city will experience a black-out?

Now suppose that it takes two plants to provide enough electricity to supply the entire city. What is the probability that the city will experience a black-out?

# Practice

Die A has orange on one face and blue on 5 faces. Die B has orange on 2 faces and blue on 4 faces. Die C has orange on 3 faces and blue on 3 faces. These are unbiased dice. If all of the dice are rolled, find the probability that exactly 2 of the 3 dice come up orange.

# Coding Challenge

Optional coding challenge:

Write an R script that simulates drawing a hand of 5 cards from a standard 52-card deck. Try to find a way to simulate many draws and estimate the probability of various poker hands (e.g., full house, straight, pair)

# Random Variables

Think of a variable $X$ that represents the outcome of an experiment.

The outcome space is called the **support** of $X$

$\longrightarrow$ E.g., if $X$ is the result of rolling a die, then the support is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Compute the probability of each outcome using a function called the **PMF**:

$$P(X = x) = f(x)$$

Here, $f(x) = 1/6$ for all values in the support.

The probabilities must add up to 1: $\sum_S f(x) = 1$